

VEX COMPONENTS POSTER

Structural · Motion · Electronics · Sensors · Fasteners · Everything in the kit

VEX IQ – 2nd Generation

Plastic construction system · Grades 5–8 · VIQRC

SPECIALTY

57 parts

- 1× Pin Tool — removes pins

1× 1× Wide Spool

3× Rubber Band Anchor

24× Medium Intake Flap
- 1× Cable Anchor

2× Silicone Rubber Band

24× Short Intake Flap

1× Rope (24× Pitch)

SPROCKETS & CHAIN

111 parts

- 4× 8-Tooth Sprocket

2× 24-Tooth Sprocket
- 4× 16-Tooth Sprocket

101× Chain Links

CONNECTORS

110 parts

- 6× Large Chassis Corner Connector

10× 2×2 Corner Connector

8× 2× Wide, 2× Offset Corner Connector

11× 2× Wide, 1×2 Corner Connector

24× Double 2× Wide, 2×2 Corner Connector

5× Double 2× Wide, 1×2 Corner Connector

8× Triple 2× Wide, 2×2 Corner Connector
- 15× Small Chassis Corner Connector

4× 2×2 Offset Corner Connector

3× 2× Wide, 1×1 Offset Corner Connector

4× 2× Wide, 3× Corner Connector

8× Double 2× Wide, 2×2 Offset Corner Connector

4× 2× Wide, 60° Omni 1×2 Corner Connector

ELECTRONICS

18 parts

- 1× Robot Brain (2nd Gen) — touchscreen, 12 ports

4× Smart Motor (2nd Gen) — built-in encoder

1× Battery Charger

4× Smart Cable 200 mm

1× Smart Cable 400 mm
- 1× Controller (2nd Gen) — wireless, joysticks + buttons

1× Robot Battery (2nd Gen) — Li-ion, rechargeable

1× 2 m USB Cable (A–C)

3× Smart Cable 300 mm

1× Smart Cable 600 mm

SHAFTS & COLLARS

125 parts

- 3× 2× Pitch Squared Shaft

2× 4× Pitch Shaft

2× 6× Pitch Squared Shaft

4× 5× Pitch Squared Plastic Shaft

4× 2× Pitch Plastic Motor Shaft

5× 1× Pitch Plastic Motor Snap Shaft

28× 0.25× Pitch Shaft Spacer
- 3× 3× Pitch Squared Shaft

2× 5× Pitch Squared Shaft

6× 4× Pitch Plastic Squared Shaft

4× 2× Pitch Motor Shaft

6× 3× Pitch Plastic Motor Shaft

32× Rubber Shaft Collar

24× Washer

PINS & STANDOFFS

201 parts

- 53× Connector Pin 1×

11× Connector Pin 1×1

4× Idler Pin

15× Standoff 2× Pitch

6× Standoff 6× Pitch

11× 0.25× Pitch Standoff Extender

8× 2× Standoff Connector
- 21× Connector Pin 2×

21× Connector Pin 0×3

13× Standoff 1× Pitch

5× Standoff 4× Pitch

11× 0.25× Pitch Standoff

13× Mini Standoff Connector

9× End Standoff Connector

PLATES

11 parts

- 2× Plate 3×3

4× Plate 4×4

4× Plate 4×8

1× 2×8 Smooth Plate

SENSORS

5 parts

- 1× Touch LED — button + RGB light

1× Optical Sensor — colour + brightness

1× Inertial Sensor — gyro/accelerometer (add-on)
- 1× Bumper Switch

1× Distance Sensor — ultrasonic

Sensors plug into any Brain port — mix and match

WHEELS & TREADS

16 parts

- 2× 200 mm Travel Wheel

2× Wheel Hub (44 mm)

2× 5× Pitch Diameter Balloon Tire

1× 2×2 Centre Offset Round Lock Beam
- 2× 200 mm Travel Tire

2× 2× Wide Hub (48.5 mm)

5× Traction Links

GEARS & CRANK

23 parts

- 10× 12-Tooth Gear

6× 36-Tooth Gear

1× 1×4 Crank Arm with 23-Tooth Gear
- 4× 24-Tooth Gear

2× 48-Tooth Gear

STRUCTURAL & ANGLED BEAMS

65 parts

- 4× Beam 2×2

5× Beam 2×6

6× Beam 2×8

4× Beam 2×12

4× Beam 2×18

4× 3×5 Right Angle Beam

4× 3×4 Tee Beam

4× 45° Angle Beam
- 4× Beam 2×4

5× Beam 2×7

4× Beam 2×10

4× Beam 2×16

2× Beam 2×20

3× 2×3 Right Angle Beam

4× 3×3 30° Angle Beam

4× 3×3 60° Angle Beam

BEAMS

43 parts

- 4× Beam 1×1

7× Beam 1×4

4× Beam 1×6

4× Beam 1×10

4× Beam 1×3 Centre Lock
- 5× Beam 1×2

2× Beam 1×5

4× Beam 1×8

4× Beam 1×12

5× Beam 1×4 Low-Profile End Lock

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VEX V5

Metal construction system · Grades 9–College · V5RC

WHEELS

16 parts

4× 4 in (320 mm travel) Anti-Static Wheel

4× 4 in (320 mm travel) Omni-Directional Wheel

8× Intake Roller

GEARS & SPROCKETS

124 parts

8× 12-Tooth High Strength Metal Pinion

8× 36-Tooth High Strength Gear v2

8× 60-Tooth High Strength Gear v2

8× 84-Tooth High Strength Gear

32× High Strength Metal Gear Insert

32× Free-Spinning Gear Insert

4× 6-Tooth High Strength Sprocket

6× 12-Tooth High Strength Sprocket

6× 18-Tooth High Strength Sprocket

2× 24-Tooth High Strength Sprocket

2× 30-Tooth High Strength Sprocket

8× High Strength Shaft Adapter (1/8 in square bore)

ELECTRONICS

28 parts

1× V5 Robot Brain — colour touchscreen, 21 smart ports, 8 three-wire ports, SD slot

1× V5 Robot Radio — wireless link to Brain

1× V5 Robot Battery (Li-ion 1100 mAh)

1× Battery Charger + 180 mm battery cable

8× V5 Smart Motor — built-in optical shaft encoder

2× 36:1 Motor Gear Cartridge

3× Smart Cable 300 mm

2× Smart Cable 900 mm

1× Smart Cable 1200 mm

1× Smart Cable 1500 mm

1× USB A–Micro cable

2× Brain Attachment Flange (short + long)

1× V5 Controller — 2 joysticks, 12 buttons, LCD

Motor cartridges swap to change speed: 100 rpm (red) · 200 rpm (green) · 600 rpm (blue)

Contents: V5 Competition Super Kit 276-7040 official kit list (+ common add-ons marked)

PNEUMATICS

4 parts

1× Air Cylinders (add-on pneumatics kit)

1× Solenoid Valve (add-on)

1× Air Tanks (add-on)

1× Tubing & Fittings (add-on)

Pneumatics kit is a separate V5 add-on — uses a 3-wire port + solenoid

SHAFTS & BEARINGS

820 parts

2× Shaft 2 in (4× pitch)

4× Shaft 3 in (6× pitch)

2× Shaft 4 in (8× pitch)

16× Shaft 12 in (24× pitch)

4× Shaft 12 in High Strength (24× pitch)

20× High Strength Shaft Bearing

12× Pillow Block Bearing

40× Bearing Flat

20× High Strength Clamping Shaft Collar

96× Star Drive Shaft Collar

30× Rubber Shaft Collar

8× Drive Shaft Bar Lock

32× #8-32 Setscrew

60× HS Shaft Spacer 0.0625 in

20× HS Shaft Spacer 0.125 in

20× HS Shaft Spacer 0.25 in

80× Nylon Spacer 0.125 in

70× Nylon Spacer 0.375 in

20× Standoff 0.5 in

20× Standoff 1 in

30× Standoff 0.75 in

4× Standoff 3 in

40× Standoff 2 in

SENSORS

14 parts

1× AI Vision Sensor — camera, colour + AI detection

4× Bumper Switch v2

1× Distance Sensor (add-on) — ultrasonic

1× Optical Sensor (add-on) — colour, light, gesture

1× Inertial Sensor (add-on) — gyro/accelerometer

1× Touch LED (add-on)

1× 3-wire: Limit Switch (add-on)

1× Rotation Sensor (add-on) — shaft position

1× 3-wire: Potentiometer (add-on)

1× 3-wire: Line Tracker (add-on)

1× 3-wire: Ultrasonic Rangefinder (add-on)

TOOLS

685 parts

4× T15 Star Screwdriver

2× T15 Star Key

2× T8 Star Key

6× Open-End Wrench

600× Zip Tie 4 in

50× Zip Tie 11 in

20× #64 Rubber Band

1× Anti-Slip Mat 12×15 in

CHAIN & TREADS

974 parts

700× High Strength Chain Links

100× Chain Attachment Links

50× Tank Tread Traction Links

60× Conveyor Belt Base Links

24× Short Conveyor Belt Insert

20× Medium Conveyor Belt Insert

20× Tall Conveyor Belt Insert

FASTENERS

1925 parts

175× #8-32 x 0.25 in Locking Star Drive Screw

125× #8-32 x 0.50 in Locking Star Drive Screw

250× #8-32 x 0.375 in Star Drive Screw

225× #8-32 x 0.50 in Star Drive Screw

100× #8-32 x 0.875 in Star Drive Screw

75× #8-32 x 1.00 in Star Drive Screw

50× #8-32 x 1.25 in Star Drive Screw

50× #8-32 x 1.50 in Star Drive Screw

25× #8-32 x 1.75 in Star Drive Screw

10× #8-32 x 2.00 in Star Drive Screw

700× #8-32 Nylock Nut

100× #8-32 Hex Nut

20× 1-Post Hex Nut Retainer

20× 1-Post Hex Nut Retainer with Bearing Flat

STRUCTURE

60 parts

2× Aluminium C-Channel 1×2×1×25

24× Aluminium C-Channel 1×2×1×35

12× Aluminium C-Channel 1×3×1×35

6× Aluminium Plate 5×25

4× Aluminium Angle 2×2×25

4× Aluminium Angle 1×1×35

4× 90° Gusset Plate

4× 90° Gusset Angle